

Understanding Pressure: Atmospheric and Fluid

A Fun STEM Adventure for Grade 6

1. What Is Pressure?

Pressure is the **force pushing on an area**. Think of it like this — when you press your hand on a table, you are creating pressure! The more force you use, the more pressure you create.

Key Facts:

- Pressure depends on how much force is applied.
- Pressure can happen in air, water, and other fluids.

■ **Remember:** More force on the same area = Greater pressure!

2. Meet Atmospheric Pressure

Did you know that **air has weight**? The atmosphere surrounding Earth is made up of tiny air particles that press down on everything, including us! This invisible force pushing on us from all directions is called **atmospheric pressure**.

It Presses Down on Us	We Don't Feel It	It Affects Weather & Breathing
Air has weight and pushes down on us and everything around us, like an invisible blanket covering the Earth.	We don't notice the pressure because it pushes equally from all directions, balancing out perfectly!	Changes in atmospheric pressure can bring rain or sunshine, and even affect how easily we breathe.

3. Experiments with Atmospheric Pressure

Try these fun experiments at home or in class!

■ **Crushing Can Demo**

■ **Inverted Glass Trick**

Heat an empty can, then quickly flip it into cold water — watch it crush! The steam inside condenses, creating low pressure, and atmospheric pressure crushes the can.

Fill a glass with water, cover with a card, and flip it upside down — the water stays inside! Atmospheric pressure pushes up on the card, holding the water in.

4. What Is Fluid Pressure?

Fluid pressure is the pressure inside a liquid or gas that pushes in all directions at the same time.

Key Points to Remember:

- Fluid pressure pushes **equally in all directions**.
- It **increases with depth** in liquids.

Fluid Pressure in Everyday Life:

■ Dam Power	■ Blood Flow	■ Tire Pressure
Water pushes against dam walls with tremendous force, helping generate electricity!	Blood pressure pushes blood through our veins, delivering oxygen to every part of our body.	Air pressure inside tires keeps them firm and helps vehicles move smoothly.

5. Experiment: Water Pressure Demo

Try this fun experiment at home! You will need: a plastic bottle, tape, and water.

Step 1	Take a plastic bottle and poke 3 holes at different heights.
Step 2	Cover the holes with tape and fill the bottle with water.
Step 3	Remove the tape and watch what happens!

■ **Notice:** Water shoots out faster from the bottom hole because pressure increases with depth!

6. Why Does Pressure Increase with Depth?

■ **At the Top of the Water**

■ **At the Bottom of the Water**

- Less water above you
- Less weight pushing down
- Lower pressure on your body

Think of it like a stack of books — one book is light!

- More water above you
- More weight pushing down
- Higher pressure all around

More books stacked = heavier load!

7. Atmospheric vs. Fluid Pressure

Atmospheric Pressure	Fluid Pressure
● Pushes down from the air above us ● Created by the weight of air in the atmosphere ● Affects weather patterns and storms ● Same pressure all around us at sea level ● Measured using a barometer	● Pushes equally in all directions ● Found inside liquids and gases ● Increases with depth underwater ● More fluid above = more pressure below

8. Why Is Pressure Important?

■ Weather Patterns	■ Submarines & Balloons	♥ ■ Our Body Functions
Atmospheric pressure changes help us predict rain, storms, and sunny days. Weather forecasters use pressure readings every day!	Submarines use pressure to dive deep underwater. Hot air balloons rise because warm air creates different pressure. Science makes them work!	Blood pressure keeps our heart pumping blood through our body. Pressure helps us breathe air into our lungs every moment!

9. Quick Recap: Key Points

■ What Is Pressure?	■ Atmospheric Pressure	■ Fluid Pressure
Pressure is the force pushing on an area. The more force applied, the greater the pressure!	Comes from the weight of air all around us. Air has weight and pushes on everything!	Pushes in all directions inside liquids and gases. Pressure increases as you go deeper!

10. Fun Quiz Time! ■

Answer these questions to test your knowledge!

Q1

What pushes on us from the air?

Answer: _____

Q2

What happens to water pressure as you go deeper?

Answer: _____

Q3

Does fluid pressure push equally in all directions?

Answer: _____

Q4

Name one example of atmospheric pressure in action.

Answer: _____

Pressure is everywhere — from the air above us to the water in our bottles. Keep exploring, keep discovering! ■